

Inducing Cognitive Control Shifts in the AX-CPT

Several prior studies have successfully induced participants to preferentially engage proactive control in the AX-CPT by providing explicit training with a proactive strategy (Paxton et al., 2006; Braver et al., 2009; Edwards et al., 2010). The training involved specific labeling and utilization of contextual information from the cue, in order to anticipate and prepare the probe response ahead of time. Participants in the strategy training condition demonstrated a shift toward proactive control, indicated by improved performance on BX trials and decreased performance on AY trials (Paxton et al., 2006; Braver et al., 2009; Edwards et al., 2010), and by an increase in sustained activation in the dorsolateral PFC during the delay period of the task (Braver et al., 2009; Edwards et al., 2010). However, this method has only been tested in small samples from populations that are known to have cognitive control and AX-CPT impairments, such as older adults (Paxton et al., 2006; Braver et al., 2009) and schizophrenic patients (Edwards et al., 2010); strategy training could conceivably be inappropriate for participants who demonstrate a higher baseline level of proactive control. Therefore, our first objective was to evaluate the validity of proactive strategy training in a large non-clinical sample of young adults.

Fewer studies have tried to induce participants to reduce utilization of proactive control. Some studies have tried to present irrelevant distractor letters during the delay period of the AX-CPT to disrupt the active maintenance of contextual information and impair proactive control in participants (Braver et al., 2001; Dreisbach, 2006; Fröber and Dreisbach, 2016). However, it is unclear whether this manipulation actually induces participants to refrain from using proactive control, or whether it merely affects performance by decreasing the effectiveness of proactive control. Another study included no-go trials in the AX-CPT, in conjunction with a penalty manipulation (Braver et al., 2009). Participants were required to withhold their response on no-go trials (which occurred with low frequency and unpredictably during probe presentation), and were assessed a monetary penalty when they failed to do so. The hypothesis was that this manipulation would deter participants from using a proactive strategy, as preparing a response in advance would elicit more errors on the no-go trials. The no-go/penalty manipulation seemed successful in discouraging a proactive strategy, in that participants performed worse on BX trials in this condition and demonstrated a reduction in sustained activity in the dorsolateral PFC during the delay period of the task. This work has important shortcomings, however. First, it was only tested in a small sample of 16 participants in the context of a neuroimaging study. Second, the no-go manipulation failed to demonstrate significant effects on most behavioral indices of performance, presumably due to the small sample size. Third, it is unclear whether monetary penalties are a necessary component of the induction; it would be useful to determine whether no-go trials are effective in the absence of incentives, especially because this would allow for interpretations that do not require

invocation of motivational mechanisms. Therefore, our second objective was to assess the validity of the no-go manipulation to bias participants away from proactive control in a larger sample of healthy young adults, without the use of monetary incentives.

In summary, the set of experiments presented here aimed to validate two methods for experimentally manipulating cognitive control in the AX-CPT. Experiment 1 tested whether strategy training can be used to increase the use of proactive control, Experiment 2 tested whether no-go trials can be used to decrease the use of proactive control, and Experiment 3 tested the combination of both manipulations in the same task. A significant feature of the experiments reported here is that they were performed at different universities, and that minor aspects of the AX-CPT (such as the presence of feedback after each trial, the precise length of the inter-stimulus interval, and the total number of trials) were varied across experiments: given that many variants of the AX-CPT have been used in past research, it was of interest to verify that our findings were not artifacts due to specific task or sample features. Procedural changes across experiments are documented in Appendix A. Relevant material for all experiments in this article (including task scripts and data files) can be accessed via the Open Science Framework platform at osf.io/6sqvw/.

EXPERIMENT 1

The aim of Experiment 1 was to validate a version of the AX-CPT that induced proactive control by extending the strategy training method developed in earlier studies to a population of non-clinical young adults. Healthy young participants were expected to demonstrate a high level of proactive control at baseline (Paxton et al., 2008; Braver et al., 2009; Edwards et al., 2010), and strategy training was expected to bolster proactive control even further.

Method

Statistical Power

The required sample size was estimated with a power analysis using G*Power 3.1 (Faul et al., 2007). Based on prior work (Paxton et al., 2006; Edwards et al., 2010), we expected effect sizes ranging between $\eta_p^2 = 0.15$ and $\eta_p^2 = 0.20$ for the effect of strategy training on measures of interest. The necessary sample size to detect an effect size of $\eta_p^2 = 0.15$ with 0.95 power was estimated to be $N = 77$.

Participants

A sample of 78 students at Princeton University completed the experiment in exchange for partial course credit or payment (\$12.00 per hour). All participants were native English speakers between the ages of 18 and 26 ($M = 19.6$ years, 30 male and 49 female). The experiment was approved by an ethics committee (Princeton University institutional review board); all participants provided written informed consent in accordance with the declaration of Helsinki.

Materials

AX-CPT

A version of the AX-CPT was constructed and implemented using E-prime software (Schneider et al., 2002). The version of the AX-CPT used in this study was based on the version used by Braver et al. (2009). Each trial began with a cue. The cue was a letter (any letter except X, K, or Y) presented in the center of the screen for 1000 ms. An unfilled inter-stimulus interval of 4000 ms followed. After the inter-stimulus interval, the probe appeared. The probe was a letter (any letter except A, K, or Y) presented in the center of the screen for 500 ms. Following the probe, a row of asterisks appeared during the 1000 ms inter-trial interval. Participants were instructed to press the target button with the middle finger of their right hand as quickly as possible whenever they observed an A cue followed by an X probe, and to press the non-target key with the index finger of the right hand as quickly as possible whenever they observed any other letter pair. Participants were instructed to respond only once they observed the second letter in the pair (i.e., the probe). Responses to the probe stimuli were recorded with a time limit of 1500 ms.

The proportions of trial types were based on those used by Richmond et al. (2015): 40% of the trials in each task block consisted of an A followed by an X (AX trials), 10% of the trials in each block consisted of an A followed by a letter other than X (pseudo-randomly selected; AY trials), 10% of the trials in each block consisted of a letter other than A (pseudo-randomly selected) followed by an X (BX trials), and 40% of the trials in each block consisted of a letter other than A (pseudo-randomly selected) followed by a letter other than X (pseudo-randomly selected; BY trials)¹. Trials within each block were presented randomly.

Data processing

Error rates and average response times (RTs; computed for correct responses only) were recorded separately for each of the four trial types (AX, AY, BX, and BY). Three additional indices reflecting the use of proactive control were also computed: the d' -context, the A-cue bias, and the Proactive Behavioral Index (PBI). The first two indices, d' -context and A-cue bias, are based on signal detection theory (Stanislaw and Todorov, 1999). The d' -context index was calculated by computing a d' index from hits on AX trials and false alarms on BX trials as $Z(H) - Z(F)$, with H representing hits on AX trials, F representing false alarms on BX trials, and Z representing the z-transform of a value. This measure reflects the ability of the participants to use contextual information from the cue to drive their answer on the probe (e.g., Barch et al., 2001). An A-cue bias measure was also calculated (Richmond et al., 2015) by computing a c criterion from hits on

AX trials and false alarms on AY trials as $1/2 * (Z[H] + Z[F])$, with H representing hits on AX trials and F representing false alarms on AY trials². This measure reflects the tendency of participants to make a target response following an A cue, independently of the identity of the probe. The third index was the PBI (Braver et al., 2009), calculated as $(AY - BX)/(AY + BX)$. This index reflects the relative balance of interference between AY and BX trials: a positive PBI reflects higher interference on AY trials, indicating proactive control, whereas a negative PBI reflects higher interference on BX trials, indicating reactive control. The PBI was computed separately for error rates (based on average error rates on AY and BX trials) and for RTs (based on average RTs on AY and BX trials); a composite PBI was also computed by averaging the PBIs obtained for error rates and RTs after standardization. The standardization was performed by using the average and standard deviation calculated over both conditions of the AX-CPT to allow for comparison between the two. In order to correct for trials where error rates were equal to zero, a log-linear correction was applied to all error rate data prior to computing the d' -context, the A-cue bias and the PBIs (as in Braver et al., 2009; see also Hautus, 1995). This correction was applied as error rate = (number of errors + 0.5)/(number of trials + 1).

These three derived indices were of interest for two major reasons. First, they have been used in a large share of research based on the AX-CPT (e.g., Cohen et al., 1999; Barch et al., 2001; Braver et al., 2001, 2005, 2009; MacDonald and Carter, 2003; Paxton et al., 2008; Edwards et al., 2010; Richmond et al., 2015). Second, they provide an efficient way of summarizing the involvement of proactive control because they combine information from multiple trial types in a single measure. For example, one condition may be associated with slowing on both AY and BX trials, but BX trials may be slowed to a greater extent than AY trials; this shift in the balance between trial types is typically interpreted as a reduced use of proactive control (e.g., Barch et al., 2003; MacDonald and Carter, 2003; Braver et al., 2005; Redick and Engle, 2011), and directly translates into a lower PBI.

Strategy training

The strategy training procedure was based on the methods used by Paxton et al. (2006). The training comprised three steps: participants were informed that an X probe was very likely to follow an A cue; they were asked to mentally prepare for a target response during the inter-stimulus interval whenever they saw an A cue, and to prepare for a non-target response otherwise; and they were trained to implement this advance preparation strategy in a series of 60 practice trials. Details of the procedure and corresponding instructions are described in Appendix A.

Procedure

Participants underwent two experimental sessions 4–14 days apart in groups of up to six participants. During the first session, participants received detailed instructions for the AX-CPT. After

¹This repartition of trial types differs from the original AX-CPT, which includes 70% of AX trials and 10% of each of the other trial types. Similar effects would be expected for the original AX-CPT, and both versions seem to usually yield comparable results. However, the modified version used by Richmond et al. (2015) has the desirable effect of equating the overall frequencies of A cues, B cues, X probes, and Y probes, thus removing possible confounds associated with first-order frequency effects. This design also maintains expectancy effects, since it leaves the conditional probability of target responses relatively unaffected (e.g., 80% of A cues are followed by an X in the modified version, as opposed to 87.5% in the original AX-CPT).

²This formula for response bias actually follows Snodgrass and Corwin (1988) rather than the more usual formula ($-1/2 * (Z[H] + Z[F])$). This is to emphasize that more liberal target response biases would be associated with higher positive numbers.

receiving instructions, they viewed a demonstration of multiple trial types in which the correct response was displayed on the screen. Following the demonstration, participants completed 10 practice trials. The experimenter observed participants during the practice trials to ensure that they were completing the task correctly. All participants were able to understand the instructions and perform the task correctly during the practice. The participants then completed four blocks of 50 trials of the AX-CPT for a total of 200 trials (80 AX, 20 AY, 20 BX, 80 BY). The data from this first session were used as a baseline. During the second session, participants completed a first block of 50 trials identical to the first session, then received the strategy training and completed the corresponding practice trials. They then completed three blocks of 50 trials of the AX-CPT for a total of 150 trials; these trials were identical to the baseline version of the AX-CPT completed during the first session. The order of the two testing sessions could not be counterbalanced because a strategy training performed first would presumably have transferred to a baseline session performed second.

Results

Two participants were excluded from the sample because they had very high error rates (one had >40% errors on AX trials, the other had 100% errors on AY trials), suggesting that they either failed to understand the instructions or could not perform the task successfully. The final sample included 76 participants. Descriptive statistics for the AX-CPT as a function of task condition are presented in **Table 1** and illustrated in **Figures 1** and **2**.

Relative to the baseline AX-CPT, participants in the strategy training condition were expected to demonstrate greater proactive control, as indicated by worse AY performance, better BX performance, and higher PBI, d' -context, and A-cue bias values. All analyses were conducted using the general linear model. For error rates, the main effect of task condition

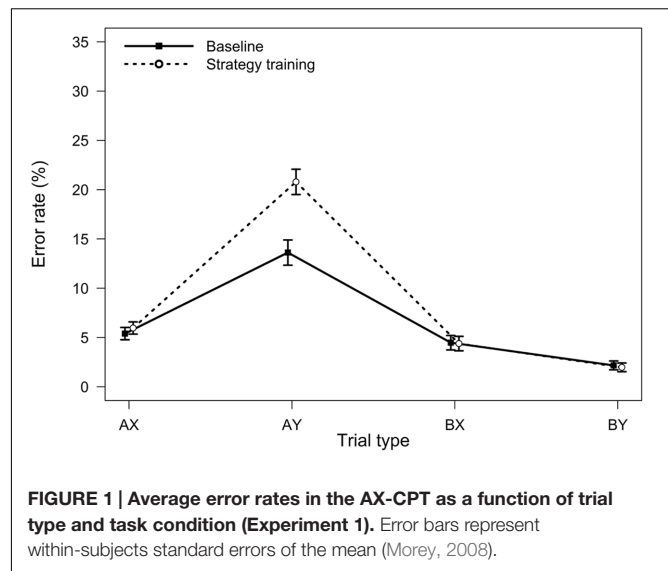


FIGURE 1 | Average error rates in the AX-CPT as a function of trial type and task condition (Experiment 1). Error bars represent within-subjects standard errors of the mean (Morey, 2008).

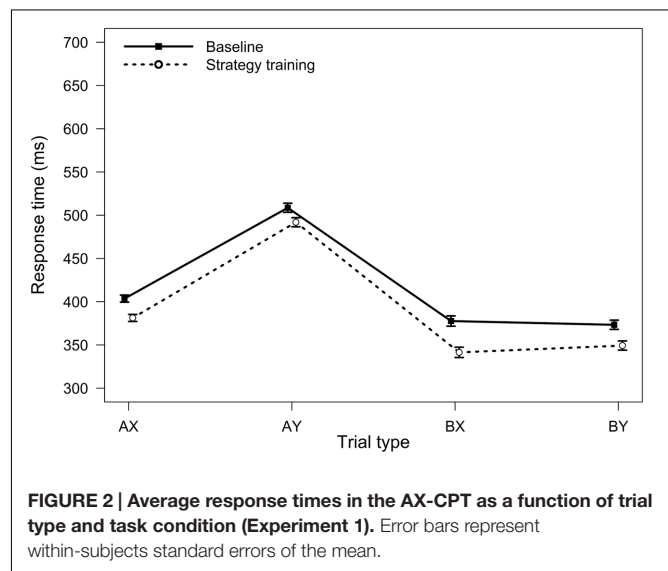


FIGURE 2 | Average response times in the AX-CPT as a function of trial type and task condition (Experiment 1). Error bars represent within-subjects standard errors of the mean.

TABLE 1 | Descriptive statistics for the AX-CPT as a function of task condition (Experiment 1).

Dependent variable	Trial type	Baseline condition	Strategy training condition
Average error rate	AX	0.054 (0.055)	0.060 (0.062)
	AY	0.136 (0.103)	0.208 (0.155)
	BX	0.045 (0.066)	0.044 (0.077)
	BY	0.022 (0.042)	0.020 (0.039)
Average RT	AX	404 (65)	381 (63)
	AY	509 (80)	492 (85)
	BX	378 (96)	341 (78)
	BY	373 (72)	349 (86)
PBI-errors		0.376 (0.365)	0.503 (0.356)
PBI-RTs		0.155 (0.074)	0.184 (0.055)
d' -context		3.34 (0.67)	3.39 (0.86)
A-cue bias		0.297 (0.288)	0.410 (0.345)

Average values with standard deviations in parentheses. PBI-errors, PBI calculated on errors; PBI-RT, PBI calculated on median response times.

(baseline or strategy training) was significant, $F(1,75) = 5.76$, $MSE = 0.009$, $p = 0.019$, $\eta_p^2 = 0.07$, indicating higher error rates after strategy training. Importantly, the two-way interaction between task and trial type was significant, $F(3,225) = 12.10$, $MSE = 0.004$, $p < 0.001$, $\eta_p^2 = 0.14$, indicating that the pattern of performance differed in the baseline and the strategy training conditions (**Figure 1**). As predicted, follow-up t -tests revealed that participants made more errors on AY trials after strategy training, $t(75) = -3.95$, $p < 0.001$, $\eta_p^2 = 0.17$, indicating an increased tendency to use proactive control in this condition. Surprisingly, however, error rates on BX trials were unaltered, $t(75) = 0.08$, $p = 0.933$, $\eta_p^2 = 0.00$.

For RTs, the main effect of task condition was significant, $F(1,75) = 17.02$, $MSE = 5512$, $p < 0.001$, $\eta_p^2 = 0.18$, reflecting a general speeding of RTs after strategy training. The two-way interaction between trial type and task condition was also significant, $F(3,225) = 2.82$, $MSE = 909$, $p = 0.040$, $\eta_p^2 = 0.04$,